TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-Performance FPGA IP Core Development for KpqC-Based Secure Embedded Devices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		**Information & Communication**	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Network System Security	Software	Embedded Software
Concept of Proposed Technology	Recent advancements in quantum computing have raised significant concerns regarding the security of traditional public-key cryptographic systems such as RSA and ECC. As quantum computers with sufficient computational power emerge, these algorithms may be rendered ineffective within a short period of time. In response, global standardization efforts for Post-Quantum Cryptography (PQC) are being led by NIST in the United States. In Korea, the Korea Internet & Security Agency (KISA) has established a national PQC initiative, referred to as KpqC, through which it has conducted research, testing, verification, and technology development. These efforts have resulted in the selection of candidate algorithms for standardization and the continued establishment of a foundation for transitioning to quantum-resistant cryptographic systems at the national level. The KpqC initiative has finalized a set of public-key encryption/key encapsulation (PKE/KEM) and digital signature algorithms. However, FPGA-based implementations of these algorithms have not			

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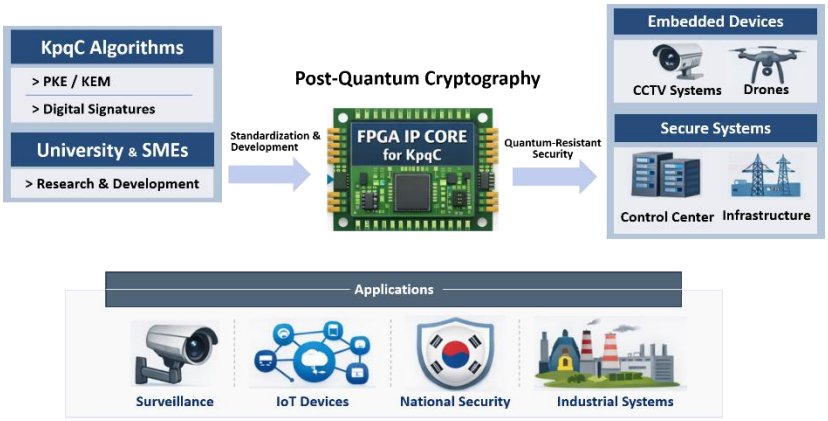
	yet been realized. Consequently, there is a critical need to develop FPGA-based cryptographic modules capable of supporting KpqC algorithms across a wide range of environments, from low-power, small-footprint embedded devices to high-performance, large-scale secure systems. Small and medium-sized enterprises (SMEs) face considerable challenges in this transition. The analysis of complex KpqC algorithms and the efficient utilization of FPGA resources for high-performance, optimized IP-core development require substantial technical expertise and infrastructure, which are often beyond the human and material capacities of SMEs. The proposed project seeks to develop an FPGA-based IP-core cryptographic module implementing KpqC algorithms, with the objective of addressing the latency and power-consumption limitations inherent in software-based implementations on embedded hardware. By enabling real-time, low-power, and highly reliable secure communication on resource-constrained, compact, and lightweight embedded platforms—such as CCTV systems and unmanned aerial vehicles—the technology will support the practical deployment of KpqC in critical national security and surveillance infrastructures. This advancement is expected to accelerate the broader transition of related industries toward quantum-resistant cryptographic systems.  [Conceptual Block Diagram of the FPGA-Based KpqC Cryptographic Module]
Rationale for Joint R&D Project	KpqC algorithms, as next-generation post-quantum cryptographic technologies, require both strong security guarantees and substantial mathematical complexity. Implementing these algorithms in FPGA-based hardware logic demands advanced research and development, including detailed algorithmic structure analysis, hardware-oriented redesign, and performance and resource optimization. The co-design process—linking the theoretical properties of cryptographic algorithms with the architectural constraints of FPGAs—introduces technical and temporal challenges that exceed the capacity of a single institution to address effectively. A joint R&D framework is therefore essential. Universities possess the specialized research capabilities and theoretical expertise required for in-depth algorithm analysis and FPGA logic design, enabling the development of core technologies and raising the overall technical readiness level. Small and medium-sized enterprises (SMEs), in turn, contribute practical industrial insight by translating these research outputs into FPGA-based cryptographic modules that meet real-world requirements for performance, power efficiency, cost, and system integration. SMEs also play a critical role in

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	validating the feasibility and applicability of the developed technologies within operational environments. This collaborative structure ensures that research outcomes do not remain confined to academic theory but are transformed into deployable, industry-relevant technological assets. By combining the complementary strengths of universities and SMEs, the joint R&D approach accelerates technology transfer, enhances industrial competitiveness, and supports the establishment of domestic technological independence. Furthermore, KpqC FPGA IP cores can be extended through wrapper modules for integration into diverse security systems, promoting broader adoption and technological advancement across SME-driven industries.
Key Words	1. Post-Quantum Cryptography (PQC) 2. KpqC (Korean Post-Quantum Cryptography) 3. FPGA IP Core 4. Embedded Security 5. FPGA-based Cryptographic Implementation 6. High-performance Low-power Cryptographic Accelerator

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (8)
R&D Objectives	The objective of this R&D project is to develop FPGA-based hardware implementations of KpqC algorithms and to validate their practical applicability in real-world security environments. Through systematic, stepwise development—from algorithm analysis and hardware-optimized design to FPGA implementation, verification, and industrial deployment—the project aims to: • Establish core FPGA IP-core technologies for KpqC algorithms that can be adopted by SMEs. • Demonstrate high-performance, low-power, and reliable post-quantum cryptographic operations on embedded and resource-constrained systems. • Enhance the technology readiness level (TRL) of KpqC-based hardware security solutions. • Facilitate technology transfer and commercialization through industry–academia collaboration, strengthening the competitiveness and self-reliance of SMEs. • Support the broader transition to quantum-resistant security infrastructures across national and industrial sectors.			
R&D Target Specifications	**1. PKE/KEM FPGA Implementation** • Implement **at least one** KpqC PKE/KEM algorithm optimized for resource-constrained embedded systems. • Achieve improved **key generation** and **encryption/decryption** performance compared to software implementations on embedded processors. • Ensure a lightweight, resource-efficient FPGA architecture suitable for embedded-class devices. **2. Digital Signature FPGA Implementation** • Implement **at least one** KpqC digital signature algorithm optimized for embedded environments. • Improve **key generation**, **signing**, and **verification** performance over embedded CPU-based software implementations. • Provide a modular, low-resource FPGA design compatible with compact embedded platforms. **3. Power Consumption Requirement** • Maintain total FPGA module power consumption **below 15 W** under normal operating conditions. **4. Intellectual Property & Research Output** • **Patent applications:** ≥ 3 • **Research publications:** ≥ 2			

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R&D Plan	**Year 1 — Algorithm Analysis & FPGA Architecture Design** • Objective: - Build the foundational framework for FPGA-based implementation of KpqC algorithms and complete the baseline hardware design. • Key Activities: - Analyze mathematical structures and computational characteristics of selected KpqC algorithms. - Identify interface requirements, parallelizable operations, and potential performance bottlenecks. - Redesign algorithms and data flows for FPGA-optimized execution. - Define baseline FPGA architecture and develop performance prediction models. - Finalize design specifications reflecting compactness, low power, and cost-efficient hardware constraints. - Conduct simulation and functional verification of core components. - Develop an initial prototype of the KpqC FPGA IP Core. • Deliverables: - KpqC FPGA logic design documentation - Prototype (simulation-level) FPGA IP Core **Year 2 — FPGA Logic Implementation & Performance Optimization** • Objective: - Advance the FPGA logic implementation and achieve target performance, resource efficiency, and power constraints. • Key Activities: - Implement full FPGA logic for selected KpqC PKE/KEM and digital signature algorithms. - Apply parallel processing structures and hardware-aware optimization techniques. - Optimize resource utilization (LUTs, BRAM, DSP) and reduce power consumption. - Conduct simulation, functional verification, and iterative refinement. - Validate performance against predefined benchmarks and embedded-system requirements. • Deliverables: - Optimized and fully implemented KpqC FPGA IP Core - Performance evaluation report (latency, throughput, resource usage, power consumption) - Interim technical report and conference-level dissemination outputs **Year 3 — Demonstration, Validation & Commercialization Linkage** • Objective: - Demonstrate industrial applicability, validate system-level integration, and establish a commercialization pathway for SMEs.

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	• Key Activities: - Build a demonstration environment for FPGA-based KpqC cryptographic modules. - Conduct system integration tests and verify operational stability in real industrial settings. - Perform final performance verification and elevate the Technology Readiness Level (TRL). - Develop technology transfer strategies and IP utilization plans. - Facilitate commercialization linkage with SMEs and evaluate productization feasibility. • Deliverables - Demonstration results report - Packaged, deployment-ready KpqC FPGA IP Core - Technology transfer & commercialization plan - Final project report, publications, and patent applications

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**1. Technology Leadership and Core Algorithm Development** • Conduct comprehensive analysis of candidate KpqC algorithms and select those suitable for FPGA-based implementation. • Design the overall FPGA logic architecture required for PKE/KEM and digital signature algorithms. • Develop and verify the core computational logic of KpqC algorithms on FPGA platforms. • Optimize the FPGA IP Core for high performance, efficient resource utilization, and scalability across device classes. • Collaborate with SMEs to ensure seamless user-interface integration and validate system-level interoperability. **2. Standardization Compliance and Performance Evaluation** • Verify that implemented KpqC algorithms comply with relevant cryptographic standards and security requirements. • Evaluate algorithm usability in constrained environments, including compact, lightweight, and low-power embedded systems. • Conduct comparative performance assessments and cross-verification with software-based KpqC implementations to ensure correctness and performance gains. **3. Project Management and Knowledge Dissemination** • Establish a multi-year technology roadmap and manage annual R&D milestones and deliverables. • Prepare interim and final technical reports documenting design decisions, implementation results, and evaluation outcomes. • Disseminate research findings through international conferences, peer-reviewed publications, and collaborative workshops. • Support intellectual property generation through patent preparation and technical documentation.
Expected R&R of Korean SME Partner	**1. Development of KpqC-Based FPGA Embedded Hardware Cryptographic Modules** • Compact, lightweight, and low-power embedded hardware device development for secure communication environments. • Implementation of video and large-scale data network transmission technologies applicable to drones, CCTV, and similar systems. • Design and integration of anti-tampering and other countermeasures against external attacks. • Collaboration with the research institute to design interface logic based on the KpqC FPGA IP Core.

	• Implementation and performance optimization of block cipher acceleration logic for secure data processing. • Development of network communication protocols, including session establishment and management. • Development of security APIs required for system integration and application-level security functions. **2. Field Testing and Performance Verification** • Verification of secure communication performance for video and large-scale data transmission in real-world drone and CCTV environments. • Lead field deployment and validation activities based on prior experience supplying point-to-point encrypted communication equipment for drone and CCTV systems. • Conduct inter-device secure communication tests utilizing KpqC-based modules. • Collaborate with accredited third-party test and evaluation centers to assess quantum-resistant security performance and anti-tampering robustness. • Provide real-world demonstrations, operational feedback, and improvement recommendations for system validation. **3. Productization and Commercialization** • Define productization strategies and conduct initial market analysis targeting public-sector and industrial applications. • Establish domestic and international commercialization plans, including licensing models and security service offerings. • Design for mass production of the developed hardware and security modules, ensuring manufacturability, reliability, and cost efficiency. **4. Contribution to SME R&D Capabilities** • Secure and internalize optimized KpqC-based FPGA IP Core technology. • Strengthen in-house capabilities for compact, low-power embedded hardware security solutions. • Pursue patents related to developed technologies and commercialization outcomes. • Enhance competitiveness and commercialization readiness in public, industrial, and defense security markets.
Test and Evaluation Center	• Proposing preliminary test items based on similar evaluations. • Collaboration with the university and SME partner for finalizing the items to be tested.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Scientific and Technological Outcomes** • Secure core FPGA implementation technologies and IP for KpqC algorithms, strengthening the domestic foundation for post-quantum cryptography (PQC) hardware. • Achieve enhanced computational performance, improved resource efficiency, and reduced power consumption through optimized FPGA logic compared to software-based implementations. • Deliver scalable FPGA IP cores that can be applied across diverse security systems and embedded environments, enabling broad technological adoption. • Establish design guidelines, evaluation methodologies, and architectural standards for FPGA-based PQC implementations, supporting future research and development. **2. Industrial and Commercial Impact** • Create new business opportunities and commercialization pathways for SMEs entering the PQC-based security market. • Provide quantum-resistant cryptographic technologies applicable to public, defense, and industrial sectors, elevating overall security capabilities. • Enable reliable secure communication services in compact, lightweight, and low-power embedded hardware environments, enhancing product competitiveness. • Strengthening SMEs' technological independence and competitiveness in advanced security technologies, reducing reliance on foreign solutions. • Support SME-driven development of next-generation security products and foster the creation of new domestic and global markets. **3. Societal and Strategic Impact** • Promote sustainable growth of SMEs in emerging next-generation security technology fields, contributing to national innovation capacity. • Foster specialized talent and build a collaborative research ecosystem through strong industry–academia partnerships. • Enhance technology credibility and global competitiveness through international collaborative research and real-world demonstrations. • Support the establishment of robust quantum-resistant security infrastructures across military, industrial, and public sectors. • Strengthen long-term national cybersecurity resilience and preparedness against future quantum-enabled threats.

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6. References

[1] KpqC documents

[2] Vivado Design Suite User Guide: Designing with IP (UG896)

[3] Vivado Design Suite User Guide: Designing IP Subsystems Using IP Integrator (UG994)

[4] Introduction to Altera IP Cores

[5] Altera IP Catalog and Parameter Editor

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization